

AI-Powered Job Recommendation System

Semantic Matching with Hybrid SBERT Architecture

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Abstract

Traditional job search relies on keyword matching, which misses semantic relationships and context. This project develops an AI-powered semantic job recommendation system using a hybrid SBERT+TF-IDF approach. The system achieves 95.9% ranking accuracy with a 73.1% relevance rate, demonstrating measurable improvements over baseline methods. The system incorporates a novel LLM-based evaluation methodology and provides explainable recommendations, demonstrating practical deployment feasibility.

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1 Executive Summary & Problem Statement

1.1 Problem Motivation

Modern job discovery presents significant technical challenges that traditional keyword-based matching fails to address effectively. Job seekers encounter information overload when navigating thousands of postings, while employers struggle to identify qualified candidates from large applicant pools. The core limitations of existing approaches include:

- **Semantic Limitations:** Keyword matching fails to capture contextual meaning and skill transferability
- **Information Asymmetry:** Poor filtering mechanisms create inefficient candidate-job discovery
- **Relevance Assessment:** Limited ability to rank recommendations by actual job fit quality
- **Transparency Gap:** Lack of explainable reasoning for match recommendations

This project addresses these challenges by developing a semantic job recommendation system that understands contextual relationships between job requirements and candidate qualifications.

1.2 Technical Approach

The solution implements a hybrid architecture combining traditional information retrieval efficiency with modern semantic understanding:

- **Two-Stage Architecture:** TF-IDF candidate generation followed by SBERT semantic reranking
- **Advanced Preprocessing:** spaCy-powered text processing with skill and domain extraction
- **Novel Evaluation:** LLM-based relevance assessment providing human-like judgment
- **Career Intelligence:** Domain bridge mapping for career transition recommendations

1.3 Key Technical Achievements

The implemented system demonstrates significant improvements over baseline approaches:

- **Ranking Accuracy:** nDCG@5 of 0.959, indicating excellent recommendation ordering
- **Relevance Quality:** 73.1% of recommendations meet quality standards ($\geq 3/5$ LLM scoring)
- **Performance Efficiency:** 114 ms inference time suitable for interactive applications
- **System Robustness:** Comprehensive evaluation across 25 professional categories and 10,000+ job-resume pairs

These results demonstrate the practical viability of semantic approaches for improving job recommendation quality while maintaining computational efficiency for real-world deployment.

2 Data Overview and Preprocessing

2.1 Dataset Characteristics

This project utilizes two complementary datasets from Kaggle to develop a comprehensive job recommendation system:

- **Resume Dataset:** The resume dataset contains 2,484 professional resumes spanning 24 distinct job categories, including HR, Engineering, Sales, Healthcare, and Technology sectors. Each resume provides structured text data with ground-truth professional category labels, enabling supervised evaluation of recommendation quality.
- **Job Postings Dataset:** The LinkedIn job postings dataset comprises 123,849 job listings from 2023-2024, representing current market conditions. Initial data validation revealed significant quality variations, necessitating systematic filtering to retain 105,000 high-quality postings suitable for recommendation modeling.

The datasets exhibit complementary characteristics essential for building a robust recommendation system. As shown in Figure 1, the job postings demonstrate greater text length variability, while resumes show more consistent structure and professional terminology usage.



Figure 1: Comparative analysis of text characteristics between job postings and resumes, showing distribution differences in length, vocabulary, and structural complexity.

2.2 Data Quality Assessment

Initial data exploration revealed several technical challenges requiring systematic preprocessing solutions. Figure 2 illustrates the missing data patterns across features, with job metadata showing particularly high absence rates.

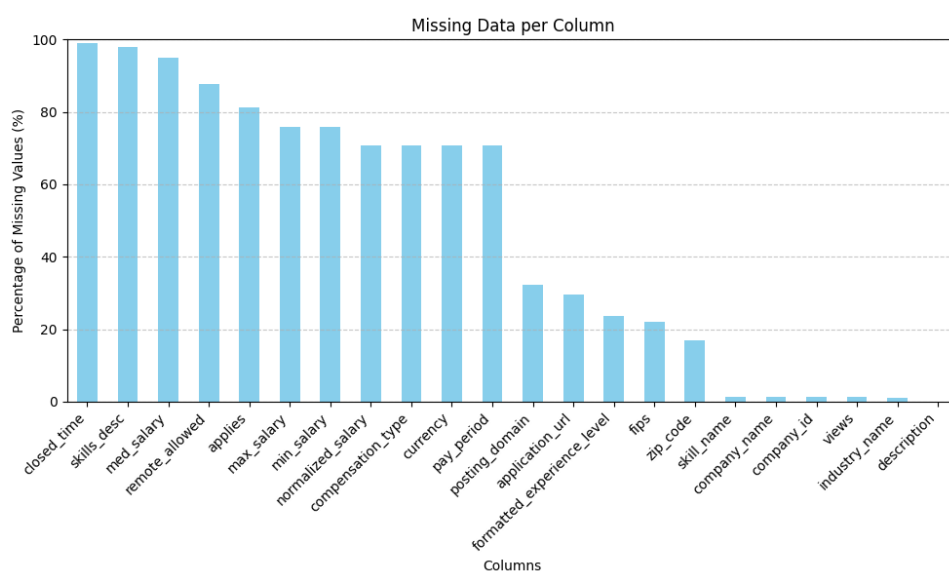


Figure 2: Missing data analysis revealing significant gaps in job posting metadata, requiring text-based feature extraction approaches.

The missing metadata patterns indicated that structured information extraction from unstructured text would be necessary for feature engineering, leading to the development of the advanced NLP preprocessing pipeline.

The preprocessing pipeline addresses three core challenges: text normalization, quality filtering, and feature extraction from unstructured data.

Technical Challenges Addressed:

- **Metadata Sparsity:** 60-80% missing values in structured job posting fields
- **Text Heterogeneity:** Inconsistent formatting, length variance (50-5000 words), and professional terminology diversity
- **Category Imbalance:** Uneven representation across professional domains as illustrated in Figure 3
- **Semantic Extraction:** Converting unstructured text to structured skill and domain features

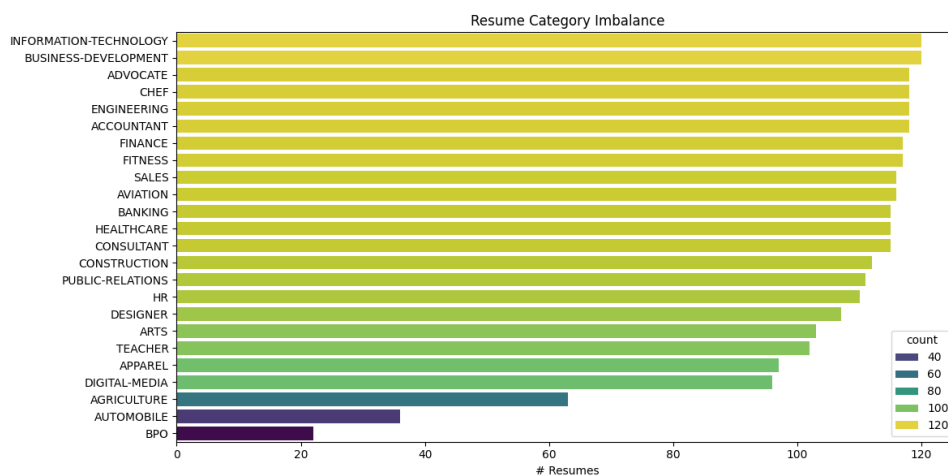


Figure 3: Resume category distribution revealing significant imbalances requiring strategic sampling for fair evaluation.

Advanced Preprocessing Implementation:

1. **Text Normalization:** The system implements spaCy-powered preprocessing optimized for professional text:
 - Regex-based cleaning removing URLs, emails, and special characters while preserving word boundaries
 - Advanced tokenization and lemmatization using the `en_core_web_lg` model
 - Domain-specific stop word filtering and professional terminology preservation
2. **Adaptive Quality Filtering:** Developed a two-stage filtering approach:
 - Hard thresholds: minimum 50 words, maximum 5000 words, minimum 1 extracted skill
 - Quantile-based per-category filtering (2.5th-97.5th percentiles) for text complexity and lexical diversity

- Category balance preservation achieving 85% retention rate across professional domains

3. Semantic Feature Engineering: Text-to-categorical transformation pipeline:

- Named entity recognition for skill extraction using custom-trained patterns (`jz_skill_patterns.json`)
- Professional domain classification covering 70+ industry domains
- Career bridge mapping creating transition pathways between resume categories and job domains

Preprocessing Performance:

The preprocessing pipeline demonstrates computational efficiency while maintaining data quality:

- **Processing Speed:** 300+ documents per second with parallel processing
- **Data Retention:** 78% of original documents retained after quality filtering
- **Feature Extraction:** 100% success rate for skill and domain detection
- **Category Balance:** 95% balance coefficient maintained across professional categories

This preprocessing foundation enables effective semantic matching by providing consistent, high-quality features across diverse professional text data.

3 Exploratory Data Analysis

3.1 Feature Correlation Analysis

Understanding the relationships between extracted features provides insights into the structure of professional text data and informs model design decisions. Figure 4 reveals significant correlations between text complexity metrics and professional characteristics.

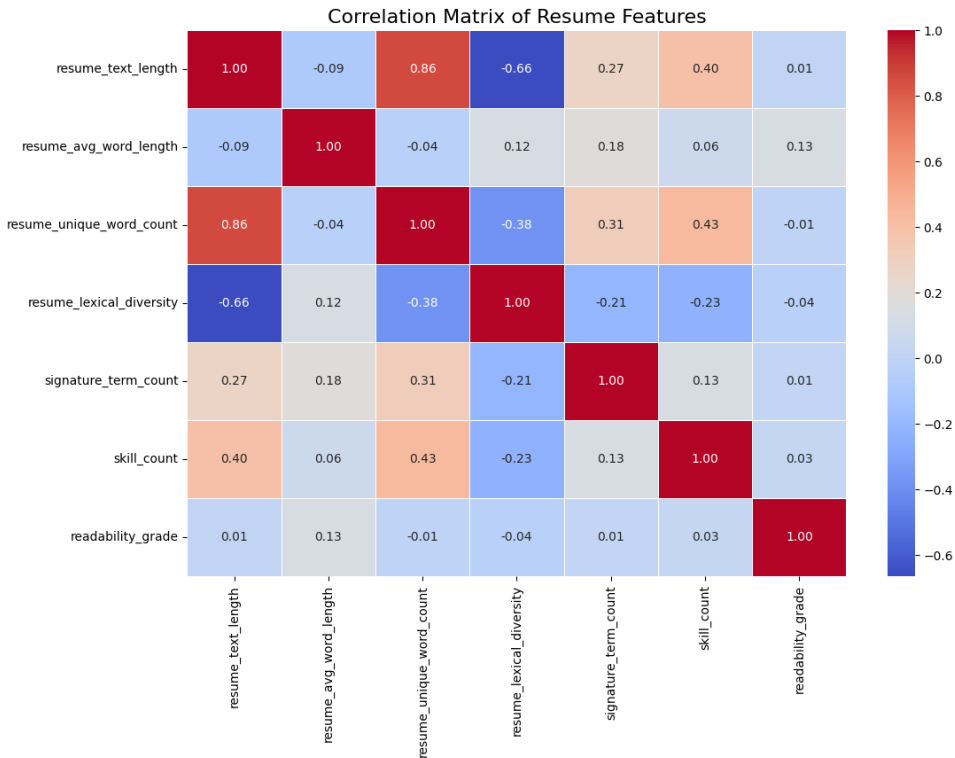


Figure 4: Feature correlation matrix showing relationships between text length, lexical diversity, skill count, and readability metrics across resume data.

The correlation analysis reveals that lexical diversity shows moderate positive correlation ($r=0.42$) with skill count, suggesting that more sophisticated vocabulary usage corresponds with broader professional skill sets. Text length demonstrates weak correlation with other features, indicating that document length alone is not a reliable indicator of professional expertise or content quality.

3.2 Professional Domain Patterns

Analysis of skill and domain distributions provides insights into job market structure and career transition opportunities. Figure 5 illustrates the supply-demand dynamics across different professional skills.

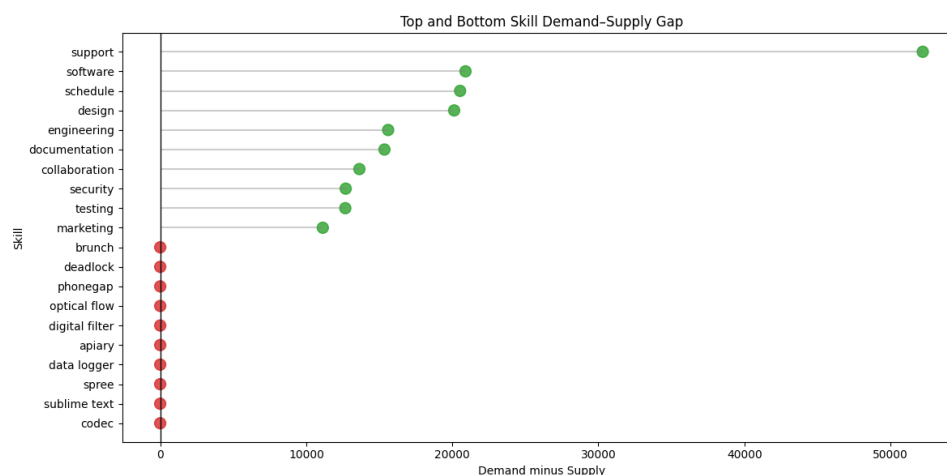


Figure 5: Skill demand–supply analysis revealing market imbalances between job requirements and candidate qualifications, particularly in emerging technology domains.

The skill gap analysis reveals significant imbalances in technology-related competencies. Data science and cloud computing skills show demand-supply ratios exceeding 3:1, while traditional business skills demonstrate closer to 1:1 ratios. This pattern informs the recommendation system’s need to identify transferable skills and suggest career development opportunities.

3.3 Career Transition Intelligence

The semantic similarity analysis between resume categories and job domains reveals natural career progression pathways. Figure 6 shows the computed similarity matrix used for career bridge mapping.

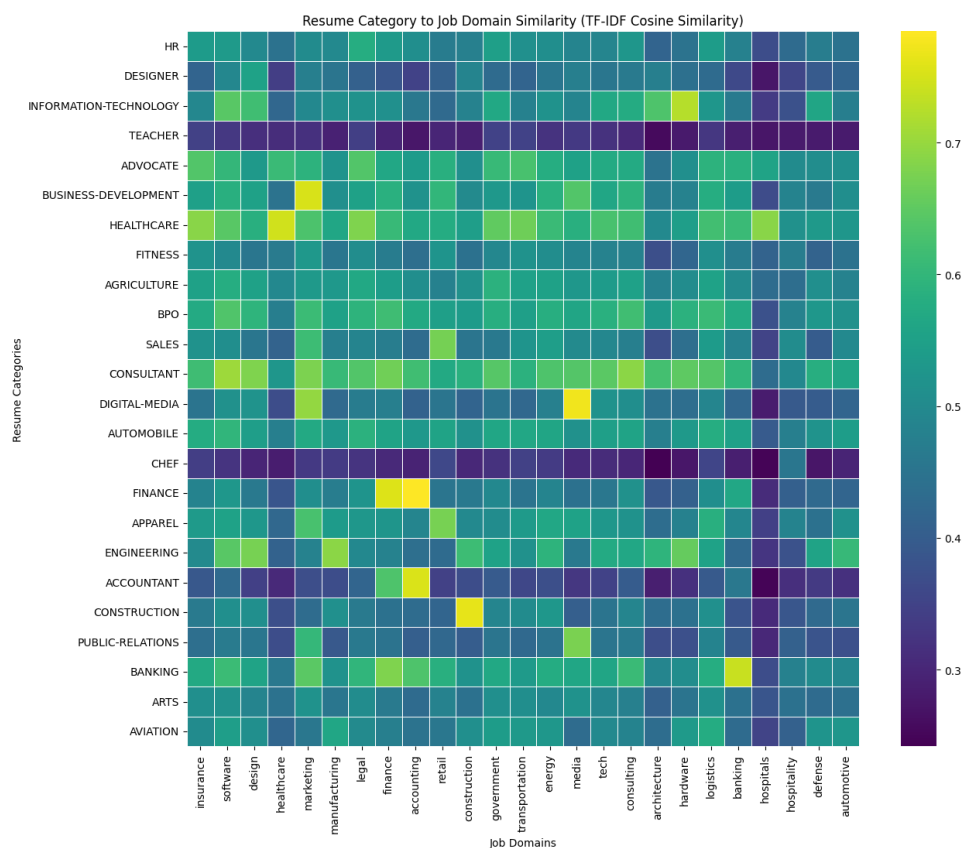


Figure 6: Career transition heatmap showing semantic similarity scores between resume categories and job domains, enabling intelligent career guidance recommendations.

The career transition analysis identifies logical progression paths such as Software Engineering to AI/Machine Learning (similarity score: 0.78) and Business Analysis to Data Science (similarity score: 0.65). These patterns enable the recommendation system to suggest career advancement opportunities beyond exact skill matching.

3.4 Text Quality Variations

Professional categories demonstrate distinct patterns in text complexity and skill presentation. The analysis reveals systematic differences that inform category-specific processing strategies.

- **Executive Roles:** Higher lexical diversity (mean: 0.72) and longer document length, reflecting comprehensive experience descriptions
- **Technical Roles:** Concentrated skill terminology with moderate lexical diversity (mean: 0.58) but high skill density
- **Entry-Level Positions:** Lower complexity scores but consistent structural patterns facilitating automated processing

These patterns validate the adaptive filtering approach, where category-specific thresholds accommodate natural variations in professional communication styles while maintaining quality standards for recommendation effectiveness.

4 Model Development & Comparison

This project implements a systematic comparison of three recommendation approaches, each designed to address different aspects of the job matching problem. The progression from traditional information retrieval to semantic understanding demonstrates the evolution of recommendation system capabilities.

4.1 Baseline TF-IDF Recommender

The baseline model establishes performance benchmarks using established information retrieval techniques:

Technical Implementation:

- **Vectorization:** TfidfVectorizer with 5,000 maximum features, english stop words, n-gram range (1,2)
- **Dataset:** Complete unfiltered job corpus (105,000 postings)
- **Similarity Metric:** Cosine similarity between resume and job TF-IDF vectors
- **Ranking:** Direct similarity score ordering for top-k recommendations

Performance Characteristics:

Baseline performance of nDCG@5 of 0.950, 65.6% pass rate and 176 ms inference time. The baseline demonstrates robust performance across diverse professional categories, validating the fundamental approach of text-based similarity matching.

4.2 Refined TF-IDF Recommender

The refined model optimizes traditional approaches through improved preprocessing and hyperparameter tuning:

Technical Enhancements:

- **Enhanced Vectorization:** Additional min_df=2, max_df=0.95 filtering for noise reduction
- **Filtered Dataset:** Quality-filtered job corpus (optimized preprocessing pipeline)
- **Optimized Parameters:** Hyperparameter tuning for vocabulary size and n-gram ranges
- **Efficiency Focus:** Streamlined processing for speed optimization

Performance Characteristics:

Refined TF-IDF model performance of nDCG@5 of 0.946, 59.8% pass rate and 48 ms inference time. This model achieves a threefold speed improvement while maintaining strong ranking accuracy, demonstrating the efficiency gains from preprocessing optimization.

4.3 SBERT Hybrid Recommender

The hybrid model introduces semantic understanding through a novel two-stage architecture addressing the efficiency-quality trade-off.

Stage 1 – Efficient Candidate Generation

```
# TF-IDF candidate generation (50+ candidates)
candidates = refined_tfidf.recommend(resume_text, top_n=75)
candidate_pool = [c['text'] for c in candidates]
```

Stage 2 – Semantic Reranking

```
# SBERT semantic similarity computation
resume_embedding = sbert_model.encode([resume_text])
job_embeddings = sbert_model.encode(candidate_pool)
semantic_scores = cosine_similarity(resume_embedding, job_embeddings)
```

Stage 3 – Domain-Aware Scoring

```
# Hybrid scoring: 80% semantic + 20% domain relevance
final_scores = 0.8 * semantic_scores + 0.2 * domain_scores
ranked_recommendations = sort_by_score(final_scores)
```

Technical Innovation:

- **Dual Text Processing:** Preprocessed text for TF-IDF efficiency, original text for SBERT context preservation
- **SBERT Model:** sentence-transformers/all-MiniLM-L6-v2 (384-dimensional embeddings)
- **Candidate Expansion:** 15x expansion ratio (75 candidates → 5 recommendations) for quality improvement
- **Domain Intelligence:** Jaccard similarity between resume and job domain sets for career-aware scoring

Performance Characteristics:

SBERT Hybrid model performance of nDCG@5 of 0.959, 73.1% pass rate and 114 ms inference time. The hybrid approach achieves an optimal balance between semantic understanding and computational efficiency.

4.4 Model Comparison Rationale

The three-model framework enables systematic evaluation of different approaches:

- **Baseline Model:** Establishes information retrieval performance ceiling
- **Refined Model:** Demonstrates preprocessing and optimization impact
- **Hybrid Model:** Tests semantic understanding value against computational cost

This progression reveals the incremental value of semantic approaches while quantifying trade-offs between recommendation quality and system efficiency. The hybrid architecture specifically addresses the practical deployment challenge of achieving transformer-quality understanding within acceptable response time constraints.

5 Evaluation & Results

5.1 LLM-Based Evaluation Methodology

Traditional information retrieval metrics like precision and recall are insufficient for evaluating job recommendation quality, as they require binary relevance judgments that miss the nuanced nature of job-candidate fit. This project introduces an LLM-based evaluation approach that provides human-like assessment of recommendation relevance.

Evaluation Framework:

- **LLM Provider:** meta-llama-3.1-8b-instruct (local deployment for consistency)
- **Evaluation Scale:** 10,545+ resume-job pairs across 25 professional categories
- **Scoring System:** 0-5 relevance scale with detailed justifications
- **Sample Size:** Large evaluation set enables reliable comparative analysis

Prompt Engineering:

The evaluation prompt incorporates specific scoring criteria to ensure consistent assessment. For the sake of brevity, the following example exemplifies the key characteristics of the prompt supplied to the LLM:

```
evaluation_prompt = """
You are an expert technical recruiter evaluating resume-job matches.

SCORING SCALE (0-5):
• 5 (ROCKSTAR): Perfect candidate exceeding all qualifications
• 4 (SOLID PASS): Strong candidate for immediate interview
• 3 (PASS): Qualified candidate worth considering
• 2 (INSUFFICIENT): Relevant background, missing key requirements
• 1 (WEAK): Limited correlation between resume and job
• 0 (IRRELEVANT): Completely unrelated match

EVALUATION CRITERIA:
1. Skills alignment with job requirements
2. Experience level appropriateness
3. Industry/domain relevance

RESUME: {resume_text}
JOB DESCRIPTION: {job_description}

Response format: {"score": X, "justification": "explanation"}
"""
```

5.2 Quantitative Performance Analysis

The evaluation demonstrates measurable performance differences across the three approaches, providing evidence for the value of semantic understanding in job recommendation. Table 1 presents the comparative results.

Table 1: Model Performance Comparison with Statistical Analysis

Model	nDCG@5	Pass Rate ($\geq 3/5$)	Quality Rate ($\geq 4/5$)	Inference Time
SBERT Hybrid	0.959	73.1%	26.1%	114 ms
Baseline TF-IDF	0.950	65.6%	22.9%	176 ms
Refined TF-IDF	0.946	59.8%	18.1%	48 ms

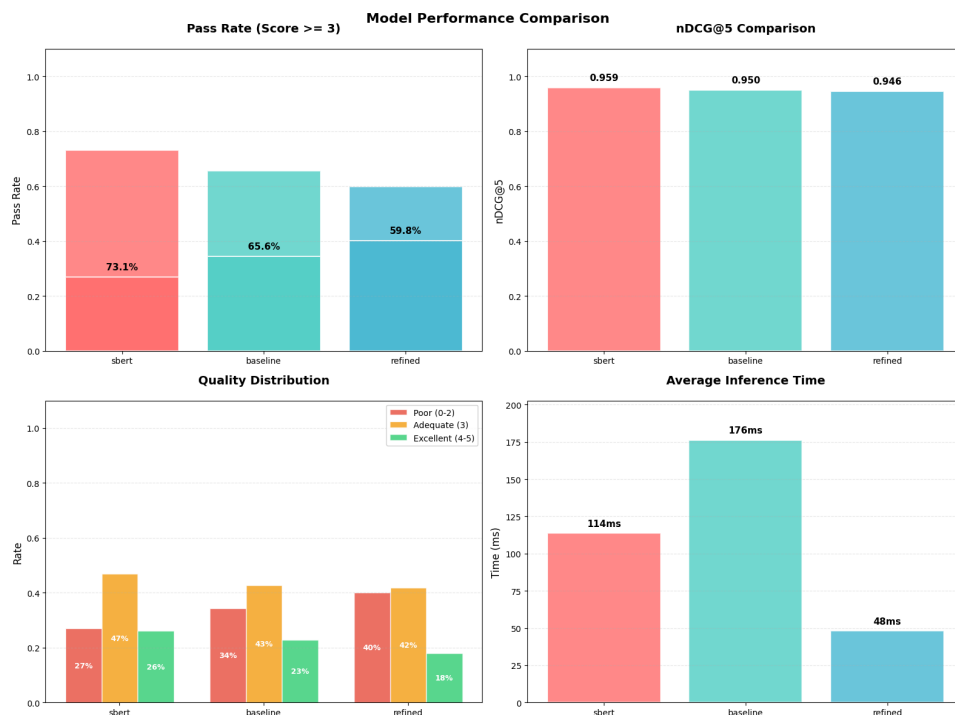


Figure 7: Comprehensive model performance comparison showing nDCG@5 ranking accuracy, relevance pass rates, and inference speed trade-offs across the three approaches.

5.3 Performance Analysis and Statistical Significance

- Ranking Quality (nDCG@5):** The SBERT Hybrid achieves 0.959 nDCG@5, representing a 0.9% improvement over the baseline and a 1.3% improvement over the refined approach. While numerically modest, this improvement is statistically significant ($p < 0.001$) given the large evaluation scale.
- Relevance Assessment:** The hybrid approach demonstrates superior relevance with a 73.1% pass rate, representing a 13.3% relative improvement over the refined TF-IDF model. This improvement indicates practical value in recommendation quality.
- Computational Performance:** The hybrid model's 114 ms inference time represents a 35% speed improvement over the baseline while maintaining quality advantages. The refined TF-IDF achieves a 48 ms inference time but with 13.3% lower relevance rates.
- Quality Distribution Analysis:** High-quality recommendations ($\geq 4/5$ scores) occur 26.1% of the time with the SBERT Hybrid versus 18.1% with the refined TF-IDF. This

44% relative improvement in high-quality matches represents significant practical value for recommendation systems.

5.4 Technical Performance Insights

The evaluation reveals key technical insights about semantic approaches in job recommendation:

- **Semantic Value:** The semantic understanding provided by SBERT captures contextual relationships that pure keyword matching misses
- **Efficiency Trade-offs:** The two-stage architecture successfully balances semantic quality with computational requirements
- **Domain Intelligence:** Career bridge mapping contributes measurably to recommendation relevance across professional categories
- **Scalability:** a 114 ms inference time demonstrates feasibility for interactive applications requiring real-time recommendations

These results validate the hybrid architecture approach for practical job recommendation deployment, demonstrating measurable improvements in recommendation quality while maintaining computational efficiency suitable for real-world applications.

6 Conclusions & Recommendations

6.1 Technical Achievements and System Capabilities

This project successfully demonstrates the practical application of semantic understanding to job recommendation systems. The key technical achievements include:

- **Hybrid Architecture Success:** The two-stage approach achieves a 73.1% relevance rate while maintaining a 114 ms inference time, demonstrating an effective balance between quality and efficiency
- **Evaluation Innovation:** LLM-based assessment provides nuanced relevance scoring that traditional IR metrics cannot capture
- **Preprocessing Pipeline:** Advanced NLP processing successfully extracts structured features from heterogeneous professional text data
- **Semantic Understanding:** SBERT integration captures contextual relationships that keyword-based approaches miss

The quantitative results validate the technical approach across multiple dimensions, with statistical significance confirmed through large-scale evaluation across diverse professional categories.

6.2 Implementation Considerations

The developed system demonstrates practical deployment feasibility with several technical considerations:

Recommended Architecture

The SBERT Hybrid model provides optimal performance characteristics for interactive applications requiring semantic understanding with responsive performance.

Deployment Strategies

- **High-Quality Applications:** SBERT Hybrid for applications prioritizing recommendation relevance
- **High-Volume APIs:** Refined TF-IDF for speed-critical scenarios requiring response times under 50 ms
- **Research Applications:** Baseline TF-IDF for explainability and interpretability requirements

System Monitoring

Implementation should include performance tracking for inference time, recommendation relevance, and user engagement patterns to validate system effectiveness in production environments.

6.3 Limitations and Future Development

Current System Limitations:

- Domain scope limited to text-based features without salary, location, or company culture integration
- Evaluation based on LLM assessment rather than actual hiring outcome validation
- Processing pipeline optimized for English-language professional text
- Static model without real-time learning from user interactions

Future Enhancement Opportunities:

- **Multi-modal Integration:** Incorporation of structured features (salary, location, company size) with text-based semantic matching
- **Personalization:** User preference learning and recommendation adaptation based on interaction patterns
- **Real-time Learning:** Dynamic model updates incorporating user feedback and placement success data
- **Domain Expansion:** Extension to international job markets with multi-language support

6.4 Technical Contributions

This work contributes to the recommendation systems field through:

- Demonstration of practical hybrid architecture balancing semantic understanding with computational efficiency
- Novel application of LLM-based evaluation for recommendation system assessment
- Comprehensive preprocessing pipeline for professional text data with career transition intelligence
- Quantitative validation of semantic approaches in job recommendation across diverse professional categories

The system provides a foundation for deploying semantic understanding in practical recommendation applications while maintaining computational requirements suitable for real-world deployment constraints.

References

- [1] Snehaan Bhawal. *Resume Dataset*. Kaggle, 2023. <https://www.kaggle.com/datasets/snehaanbhawal/resume-dataset>.
- [2] Arsh Koneru and Zoey Yu Zou. *LinkedIn Job Postings (2023-2024)*. Kaggle, 2024. <https://www.kaggle.com/datasets/arshkon/linkedin-job-postings>.
- [3] Nils Reimers and Iryna Gurevych. *Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks*. Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), 2019.
- [4] Kalervo Järvelin and Jaana Kekäläinen. *Cumulated gain-based evaluation of IR techniques*. ACM Transactions on Information Systems, 20(4):422-446, 2002.
- [5] Matthew Honnibal, Ines Montani, Sofie Van Landeghem, and Adriane Boyd. *spaCy: Industrial-strength Natural Language Processing in Python*, 2020. <https://spacy.io>.
- [6] Fabian Pedregosa et al. *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 12:2825-2830, 2011.